



HELLENIC REPUBLIC

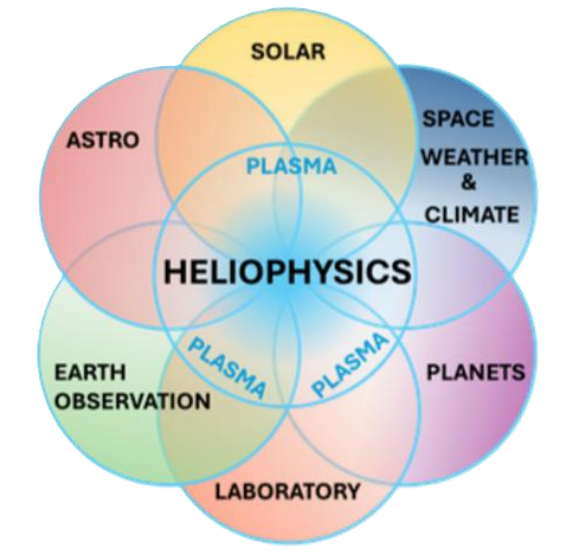
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Application of the Kennel-Petschek Limit on Electron Distributions in Jupiter's Inner Magnetosphere

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Introduction

Whistler-mode chorus waves are key drivers of acceleration and loss processes in the environment of magnetized planets. According to the Kennel-Petschek (K-P) limit, electron fluxes trapped in magnetic bottle configurations exhibit a self-limiting intensity. Once that limit is surpassed, the thermal anisotropy within the system leads to self-generation of whistler-mode chorus waves, that eventually balance the electron fluxes through pitch angle scattering.

Previous studies have proven that the K-P limit is sustained in electron populations within Earth's radiation belts, even during intense storm activity. It has also been suggested that the radiation belt environment of Jupiter may be strongly influenced by the K-P limiting processes. This study focuses on whether the Jovian differential electron fluxes are constrained from the K-P limiting processes.

Data & Methodology

- JUNO archival data were utilized for this study. Specifically, the 100-1000 keV differential electron fluxes were obtained from the JEDI instrument, and magnetic field values were obtained from the MAG instrument. The differential flux data were fitted using a Kappa distribution with a power law tail, and corrected through dedicated routines.
- A total of 100' of data were obtained from Juno's 7th Perijove. The data were obtained from the equatorial crossing ($|\lambda| < 10^\circ$), where chorus wave activity is expected to be maximized. The L shell values were 15.5 -17.0.
- A comparison of electron fluxes with the corresponding theoretical KP limit was conducted according to the methodology developed by Mauk & Fox (2010), and Olfier et al. (2022).

Results

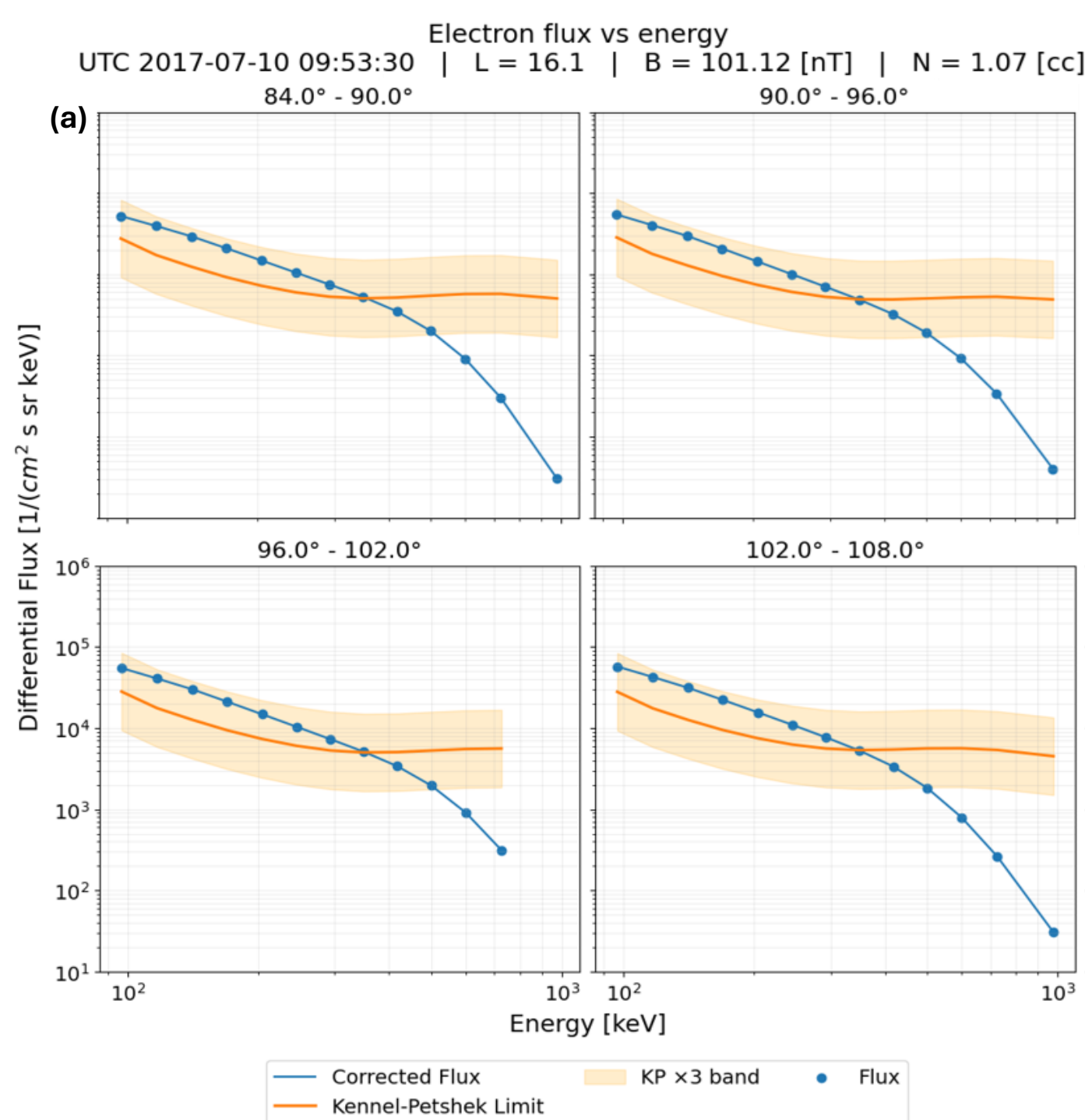


Figure 1: (a) Unidirectional electron differential fluxes and corresponding KP limits with a factor of $\times 3$ of uncertainty for pitch angles near 90° for an instance. (b) Comparative Spectrogram showcasing the evolution of 90° pitch angle flux across different energy channels per UTC time, and the corresponding K-P limit within a factor of 3.

The results indicate that near- 90° pitch angle electron fluxes within reside within a factor of $\times 3$ from the estimated K-P limit [100–550keV]. The ~ 350 keV electron fluxes act as a threshold where the fluxes intersect with the K-P limiting levels for the whole duration of the observation window.

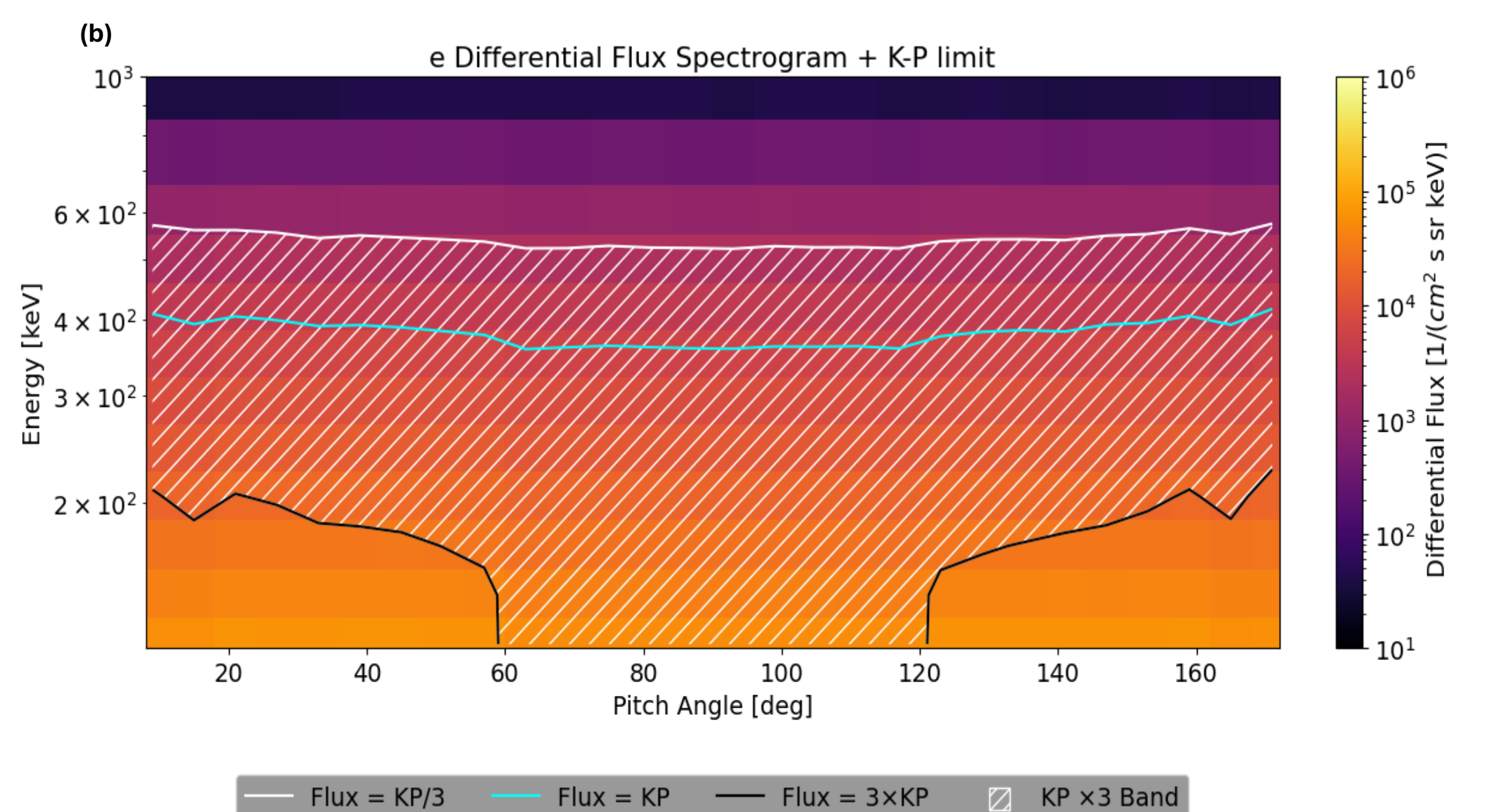
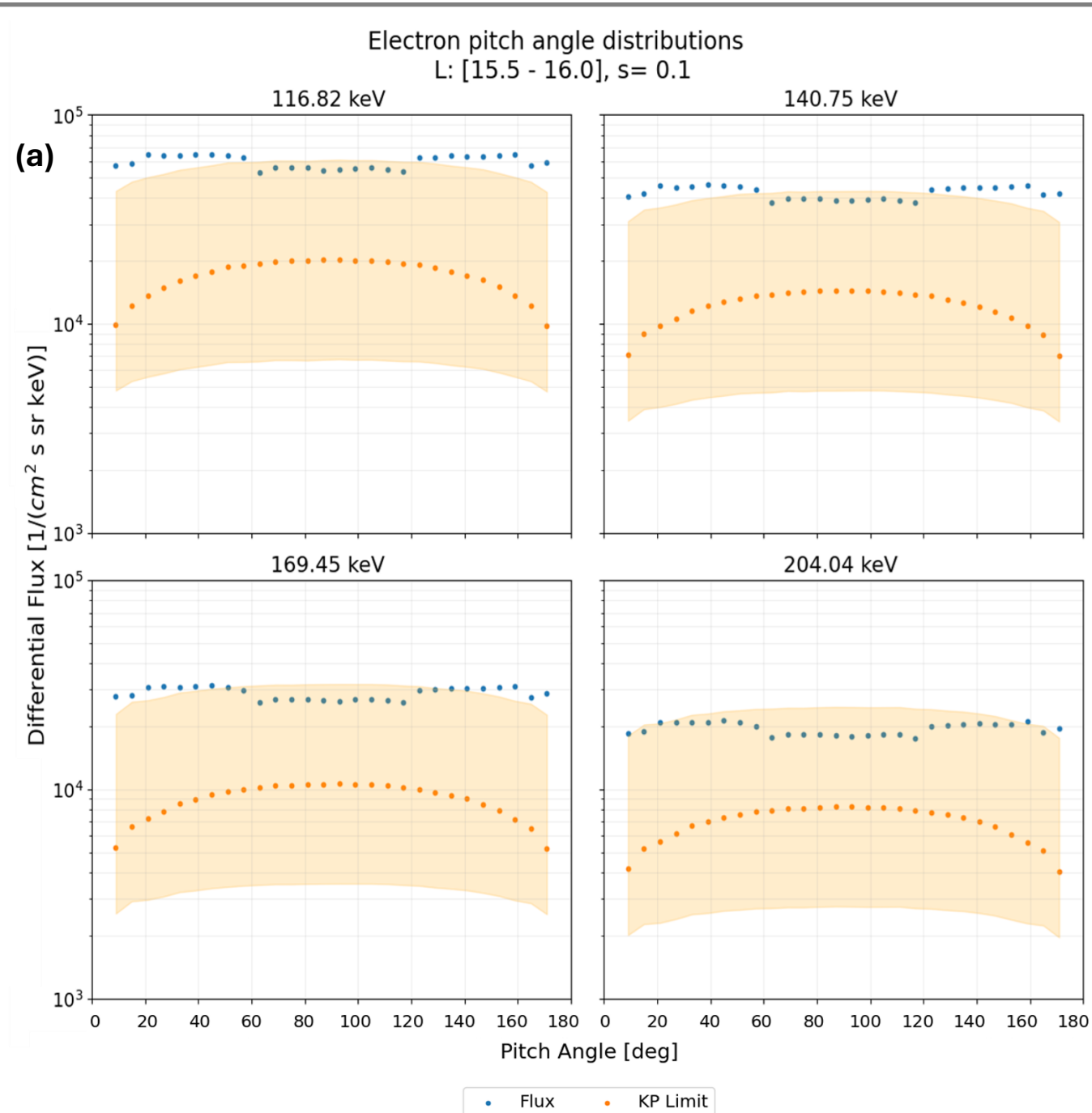
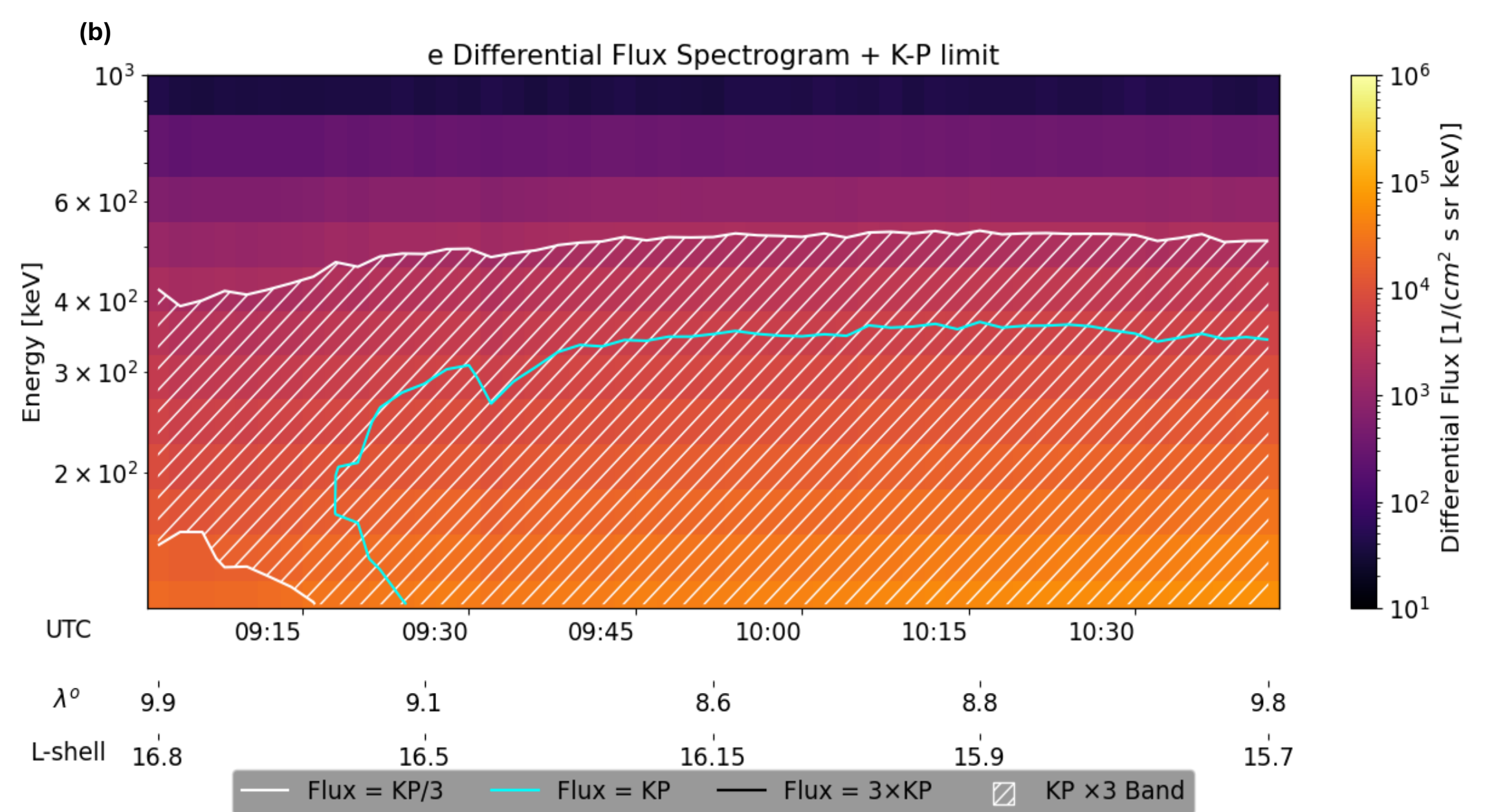


Figure 2: (a) Low energy electron pitch angle distributions and the corresponding K-P limit with a factor of $\times 3$ of uncertainty for L-shells near Ganymede's vicinity. (b) Comparative Spectrogram showcasing the evolution of pitch angle distributions of electron fluxes across different energy channels per pitch angle, and the corresponding K-P limit within a factor of $\times 3$ for L-shells near Ganymede's vicinity.

The pitch angle distributions highlight that the results are mostly consistent across different pitch angles, apart from localized exceptions at $20^\circ - 50^\circ$ and $130^\circ - 160^\circ$, where the distributions exceed the $\times 3$ uncertainty factor for short timescales.

Main Conclusions

- This study suggests that **non-relativistic electron differential fluxes near Ganymede's orbit may be strongly capped** by the self-limiting K-P processes in energy ranges of hundreds of keV.
- Relativistic energy electron fluxes reside well below** a factor of $\times 3$ of the K-P limit. This result was anticipated, as whistler-mode chorus waves do not dictate the dynamics of relativistic electrons.
- Overall, the results of this study indicate that the **Kennel & Petschek limiting processes may significantly alter the shape of the Jovian electron distributions**.

References

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